

Importance of Democratization for Virtual Reality in Education

Katrien Jacobs*

Simon Fraser University
School of Interactive Arts and Technology

ABSTRACT

Virtual Reality (VR) is a quickly growing medium that shows great potential for use in education. However, VR must first become democratized through addressing current barriers to creation. By using the guiding principles of Open Educational Resources, researchers can support the uptake of VR as an educational technology. Given the current direction of research, VR will not become viable in educational contexts until researchers prioritize its democratization.

Keywords: Virtual reality, Education, Educational technology, Open Educational Resources (OERs), Barriers to change.

1 VR RESEARCH & POTENTIAL

Virtual reality (VR) is one of the most quickly growing mediums of creation, and yet this medium has only recently started to permeate the general consciousness [1]. In recent years virtual reality has gone from a novel but underdeveloped technology, to a viable albeit cost prohibitive format, for a variety of applications. From exposure therapy to specialized surgery training, VR has slowly but steadily become a viable tool [2]. However, due in part to the complexity often required to design and implement VR experiences, this medium is still unable to be widely implemented in one of the contexts that would benefit from it the most: education. The use of VR in educational context has been discussed often and even implemented on a small-scale level. Especially in the context of the COVID-19 pandemic and necessities of remote and hybrid education, the medium gained a new level of interest from researchers and industry leaders alike [3]. However, the hypothesized and advertised potential of virtual reality for educational contexts has resulted in comparably little uptake from the corresponding institutions. This is less due to a lack of interest and more due to the currently limited number of applications suitable for a variety of educational settings.

Standard VR technology is currently far beyond the budgets of already-underfunded public school systems. While there have been efforts to make more affordable instances of VR, these lack cross-compatibility and bring in additional limitations in performance, capability, and durability. This market segmentation is profitable, especially for companies producing professional-quality VR technologies as it allows them to keep their standard headsets highly priced without limiting their target market to those with disposable income. This segmentation not only decreases the quality of affordable headsets, but also the quantity of available educational experiences [4].

As it stands, researchers are disproportionately incentivised towards improving high-end virtual reality technology performance and the fluidity of user experience [4], [5]. However, if VR is to become a viable medium for education, researchers must first focus on lowering the existing usage and creation barriers that inhibit the average educator from implementing it in their own curricula.

Virtual reality currently has incredible potential as an educational medium and the technology has developed to a point where many academic researchers have recently taken interest. Research in VR largely focuses on *end-user performance*: both increasing specifications such as visual resolution and processing power, as well as improving specific aspects of user experience, such as sense of movement through space or tactile feedback, in already created VR environments [4]. The research produced is siloed rather than holistic, and publications rarely demonstrate engagement with other subsets of the field of VR [1], [4]–[6]. While beneficial within a limited context, these incremental improvements to a specific VR engine’s capability reflect the industry-based interests of the major sources of funding for this research alongside the bias in priorities of the field overall.

This research direction implicitly prioritizes VR’s use among those with significant money and time available to invest in it. While this direction is producing tangible benefits - improving the technological capabilities of VR as a whole - that is not the most important step for VR’s growth as a medium. It is not image resolution or technical features that prevents the adoption of VR on a wider scale, but its limited use and accessibility beyond the existing clean and professional, but ultimately limited demos. Continuing with this overall research direction will not lead to greater uptake of VR across the general population.

Educators seeking to integrate VR into their classrooms are presently faced with barriers of entry that are too significant to be justified by their limited benefits. Rather than making incremental improvements to the performance and outputs of VR, researchers and industry players alike must radically shift their priorities towards a process of *democratizing* VR. This paper argues that to do so researchers should use the past successes of technology democratization as well as the “5 Rs” model used by Open Educational Resources to guide them [7]. This method shows the greatest potential for the field to identify and address the existing barriers that inhibit VR uptake in educational domains.

2 DEMOCRATIZING VR FOR EDUCATIONAL CONTEXTS

If the goal is to make VR more widely applicable to educational contexts, researchers need to focus less on the technological limits of VR, and more on the viability of general VR usage. VR must become democratized so that individual educators and school boards may utilize it without first investing in the resources necessary to overcome the current barriers to usage. For the purposes of this paper, *democratization* refers to the processes through which means of production for a technology become more broadly available to people outside industrial infrastructures and allow for playful engagement and knowledge sharing with new users [8].

If VR is to have a future beyond the standard industry uses of video games and specialized professional education, we must focus on making VR accessible to the average potential user. More

* e-mail: kjacobs@sfu.ca

specifically, we should focus on democratizing creation within Virtual Reality. Allowing anyone to *use* virtual reality is laudable, but allowing anyone to *create within* Virtual Reality is the kind of change that can allow people to develop and share their own educational VR experiences.

2.1 Current Barriers

Hardware and software's technological capabilities are not the major barrier to the wider implementation of Virtual Reality [9]. Individual educators need to be able to produce their own content in and for VR, but creation barriers prevent them from doing so. Addressing barriers to creation and adoption in educational contexts is key to democratization. Therefore, if VR is to become a viable tool within the classroom, it must first be democratized.

2.2 Democratization and Open Educational Resources

This can often depend on contextual factors such as affordability, community support, or widespread use. However, the engagement of *novice creators*, those with limited experience in the skill sets typically required to use the technology, is key to democratization across all technologies [8]. There is currently a notable lack of work done on democratizing creation within Virtual Reality [10]. This lack of VR creation democratization limits not only who can create, but what ideas can be created [8], [11]. If a system is not usable by the vast majority, it is not truly democratized [8].

A medium that has received greater uptake in educational contexts due to its explicit focus on democratization is Open Educational Resources. *Open Educational Resources (OERs)* includes a range of resources from specific learning activities to complete textbooks that are freely shared and continually improved upon by a broader community of educators [12]. This has led to a movement to build open access databases of education resources for educators and learners to use without cost. They are guided by a set of principles known as the "5 Rs": Reuse, Revise, Remix, Redistribute, and Retain [7], [13]. These principles guide educators and creators on how to approach OER content. They should *Reuse* materials that others have shared, *Revise* these materials with up-to-date information, *Remix* them to fit their curricula, *Redistribute* their revisions and remixes for others to use, and *Retain* the resources for future usage. Most pertinent to the topic of VR democratization for education are the Rs of Reuse, Remix, and Redistribute. These are well-aligned with the four key changes to technology and its context to produce democratization: *knowledge sharing*, *playfulness*, *interoperability*, and *tools for production* [8].

Specifically, they can be combined into three overarching principles to improve uptake and reduce barriers to creation. Firstly, VR creation tools should be learnable and accessible for novice users. This aligns with the key democratization process of accessible tools for production and allows anyone to engage with the resources shared; this foundation is necessary for the following two principles. Second, VR creations and learning materials should be openly shared and easily discoverable to encourage collaboration. In order for people to have access to these materials, they need to be retained and redistributed in a publicly available space (Ideally through the internet so geography is not a barrier). Finally, VR creations, tools, and assets should be cross-compatible, allowing for individuals to revise, remix, and apply them within new contexts.

In the following section, I will discuss these three principles, drawn from democratization and Open Educational Resource movements, to guide future research in virtual reality. Orienting research around these principles is crucial to allow VR applications in educational contexts to reach their full potential, rather than remain limited to those with excess resources to invest in

technology with limited use cases. To properly understand these three principles, we must explore how they have each been applied to democratization of other technologies and uptake of OERs in educational contexts.

3 THREE PRINCIPLES FOR DEMOCRATIZING VR

3.1 VR Creation Tools Should Be Learnable and Accessible for Novice Users

If VR is to become democratized, then VR creation tools must first become accessible to the average educator. It must be easy for an educator with baseline technological literacy to design an experience for their class without any prior experience in programming or design. Researchers should focus on how exactly to lower the current barriers to creation for the average user and design with novices in mind. In OERs this usually takes the form of templated study sheets, lesson plans, and activity suggestions, but specific technologies such as H5P, a creation tool for web-based OERs, actively support the design of technology-enabled learning objects [14], [15]. The teacher is able to take these resources and use them to create their own curriculum material. By helping teachers to create their own activities for use in VR, the covered topics will no longer be restricted to the most commonly taught material or approaches.

By lowering the barriers to creation for virtual reality, researchers can help to revolutionize the medium. Research should prioritize identifying the nature of these barriers as well as their impact on novice experiences. However, given the diversity of existing VR creation tools this process may not be a straightforward one. It will be important to compare the different existing tools, the ways in which these tools support novices, and the overall limits each tool can be pushed to. This comparison can help to identify the most common barriers that novice creators experience. By identifying these barriers, designers can then prototype alternatives or additional add that may assist novices in using such tools.

3.2 VR Creations and Learning Materials Should Be Openly Shared

This benefit becomes further compounded if VR modules are also sharable with other educators through a central hub. This would give educators access to the creations of others that they could use, adapt, and update to fit their own curriculum. If researchers focus on developing a central library for the modules created through their own research as well as learning activities made by educators, then both further research and the continuous development of educational experiences can more easily build upon one another. This would also allow educators to share the activities they created for their curriculum as well as for such activities to be used, updated, and reviewed for accuracy by their peers.

Many communities work to reduce the existing barriers to entry in their own way by creating and sharing many high-quality tutorials. There are numerous scholarly communities, both inside and outside educational institutions, focused on reviewing, editing, and sharing expertise around OERs. OER uptake and trustworthiness have been greatly improved through publication and database sites such as PressBooks, OER Commons, and MERLOT [16]–[18]. Some OER publishing venues have content licensing features intrinsically built in, though their reliability depends on user action [19].

OERs have also gained popularity as an alternative to the increasing cost of subscription-based online textbooks. Schools often can't afford to pay to *rent* a textbook for 4–6 months that students will only be able to access online. This cost either gets

foisted on to the students or prevents schools from using these resources. By using OERs, teachers are able to access class resources regardless of school funding and download them for use offline. Democratizing VR could have similar benefits given the increasingly commercialized state of educational technology. Rather than allowing companies to divide the market based on a school's level of funding, something affected by the affluence of students' families and local taxes, teachers would be able to freely share resources and build upon one another's work.

Currently, much VR research involves creating VR experiences to test a specific theory, after which the source files are not published or made available. This minimizes knowledge sharing and forces repeat effort, while secondarily reducing the testability of findings. Making the components of VR creations modular and shareable improves sharing of expertise and iteration on extant VR creations, accelerating evolution of the medium as a whole. To apply the benefits of open access resources to VR, researchers should collaborate on a platform to share and discover existing creations and tools, minimizing repeated work and providing credit where it is due.

3.3 VR Creations, Tools, and Assets Should Be Cross-Compatible

As it stands, there is currently an excessive number of bespoke and unmaintained "learning languages" and tools to assist in the creation of VR implementations. This demonstrates a lack of collaboration, and specifically incentives for collaboration on their approaches to VR. This issue is not unique to VR or extended reality by any means, but exacerbates the barriers inhibiting uptake of VR in other disciplines [20], [21]. It would be far more productive for prospective VR creators and current researchers to build upon a standard of generally used engines and file formats. However, this standard should not be limited to only one engine, as that restricts the approaches available to potential creators and prevents other engines from recognizing and potentially serving novice creators as an important user group.

OERs are frequently distributed with their source HTML, .docx, or PDF file formats, so interoperability is less of a technological challenge than with VR learning experiences and more focused on appropriate licensing and credit [19], [22]. By writing licenses and terms that encourage fellow educators to "remix" or create alternate modified versions of their peer's activities, people using OERs have been able to customize their content to their specific pedagogical aims, geographic context, and deliver curricula tailored to their learners' needs. Similarly, prioritizing interoperability and the ability of users to build on each other's creations for VR widens the potential for uptake and growth of individuals' contributions to the medium.

For VR, this may take the form of a general template for activities as well as support for specific common file formats [23]. Standardized formats make it easier for more novice users to produce additional user-generated content [11], [24]–[26]. Interoperability makes it easier for educators to update information, provide refinements, or tweak content to fit their own objectives [22], [27], [28]. Researchers should explore how to best implement an accessible sharing, remixing, and updating system, including how to best preview the VR experience or quickly prototype one's changes. It would also be important to determine how to best present such a format for educators to easily understand and make use of. By giving educators the ability to modify existing experiences to fit their curriculum or include updated information, it would be easier not just for the entire library to stay up-to-date but also for the material to be quickly adapted based on their target

audience. This would further help to push VR away from basic experiences to more unique and inventive uses within education.

4 CURRENT APPROACHES TO VR DEMOCRATIZATION

4.1 Industry Approaches

Even within VR, there are already efforts being taken to democratize creation, albeit within a limited scope. Two such examples of this are Dreams and CoSpaces Edu, systems which support novice users in creating within VR without any prior programming experience through the use of visual scripting. *Visual scripting* refers to the use of a visual interface to create code [29]. By using symbols and computer logic, a creator without any programming experience can "code" a function via visual scripting.

Dreams is a video game by Media Molecule for the Sony PlayStation 4 in which players can design and create their own worlds, experiences, and interactions within the game's engine [30], [31]. While not limited to VR, both the game and the system are built for players to use Sony's proprietary PSVR headset to explore their own creations as well as those others have shared online within the game. Many players even make and share VR specific creations within the community. To surmount the complicated issue of coding for virtual reality, Media Molecule implemented a flow diagram-based visual scripting tool. This allows users to drag and drop a series of individual functions like switches (logic gates) and then draw lines between them to create their indented interactions. Additionally, assets are already available to be dragged and dropped, removing the necessity for expertise in 3D modelling, level design, and particularities of physics within VR engines.

CoSpaces Edu is an engine targeted towards schools and student workshops [32]–[34]. The engine focuses on helping children create and code within a 3D engine. These creations can then be explored in either virtual reality or augmented reality. CoSpaces Edu is set up specifically for group use with a teacher overseeing the overarching system and the learners (often young children) each given access to their own space to create within. Teachers can also set up lessons and activities within the engine so they can manage classes, facilitate student collaboration, and oversee their work. CoSpaces Edu also offers its own limited programming approach with block-based visual scripting. Within this, students can connect blocks of code to create specific functions or interactions.

However, while both of these examples help make VR creation more accessible to younger and less specialized audiences, neither one truly helps democratize VR. CoSpaces Edu runs on a yearly subscription fee for both the admin and each student account they have in their class. This engine's capabilities are quite limited due to the use of block coding which relies on the needed functions already having been programmed in. Dreams also suffers in terms of financial accessibility as it is currently exclusive to the PlayStation 4. Therefore, one must own a PlayStation 4, the separately bought PlayStation VR headset and camera, and Dreams itself. While Dreams alone may be inexpensive, the cost of hardware for each student to use synchronously in a classroom setting can quickly become prohibitive.

4.2 Academic Approaches

Further compounding this issue is academia's priorities as expressed through their research output. Currently, many labs focus on VR-related research in an attempt to further develop this medium. However, their research is often focused towards areas that, while beneficial to the development of VR and (eventually) our experiences within it, research does less to benefit the general

populace and more to benefit a specific few. It is not uncommon for a VR lab to regularly work with industry partners and to focus on proprietary technology both in terms of software and hardware [5]. While beneficial in terms of receiving funding and helping to push the limits of VR technical capabilities, many researchers end up disseminating research that helps corporations more than society as a whole. By making their own specialized engines or by working with an industry-funded focus, this in turn inhibits VR's development as a medium by reinforcing the barriers to creation that limit diverse perspectives.

In order to make VR creation tools learnable and accessible for novice users, researchers should focus on creating modular plugins for a variety of popular existing VR creation engines rather than making specialized programs or tools that are never publicly released. To support the sharing of VR content, researchers should create an accessible library of such resources and past creations for others to use. In terms of asset cross-compatibility, more emphasis should be placed on the already modular nature of VR engines, many of which are built to allow for user-generated integrations. Focusing on modular additions rather than complete rebuilds would reduce barriers for researchers to build upon one another. This would also support creators' use of these resources, making creation engines such as Dreams and CoSpaces Edu more accessible to the general audience. If our goal in expanding the use of VR in educational contexts is to further our collective understanding, then we should work to further our collective abilities. By maintaining the same course, we continue to solely enrich the capabilities of industry players and limit access for those who cannot afford their specific products.

5 CONCLUSION

In conclusion, unless virtual reality becomes democratized, its potential in educational contexts cannot be achieved. For this to happen, researchers must decide where their priorities lie in terms of VR development. Do they care more about pushing the limits of virtual reality in niche contexts, or about making it accessible for use in educational contexts? It is not wrong for some researchers to remain committed to the status quo. However, unless there are those willing to diversify virtual reality research to focus on *accessible* creation, educational uses of VR will be available only to the most well-funded schools. Enabling educators to create their own interactive, immersive environments is necessary for educational uses of VR to develop from their current barrier-laden state towards their potential. Virtual reality could be a powerful tool that helps educators reach more learners and meet the needs of underserved students. Providing tools for educators to engage their students using emerging technologies can lead to incredible creations. Furthermore, by helping educators connect with one another, they can assist with knowledge transfer and the adaptation of VR learning experiences for more diverse contexts. Therefore, if the goal is for VR to become a viable tool not only for education but for a greater diversity of situations, it must first be democratized. By actively choosing not to facilitate these processes, we are not only limiting access to VR as a whole but also limiting the potential of VR as a creation medium.

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